

MAUSAM 2.0

Next-Gen Personalized Weather System & Comprehensive Code Explanation

Problem Statement ID: 26076

Ministry: Ministry of Earth Sciences (MoES)

Department: India Meteorological Department (IMD)

Theme: Smart Automation

Version: 2.1.0 (Pan-India & Global Coverage)

1. Executive Summary & Problem Background

The standard '**Mausam**' mobile application by the India Meteorological Department provides extensive radar, satellite, and district-level forecasts, but serves a generic one-size-fits-all screen. **Problem Statement ID 26076** mandates the creation of a **Personalized Homepage System** that transforms raw atmospheric data into hyper-targeted, actionable intelligence across 8 specialized user personas and dynamic lifecycle routines.

Key Innovations Delivered:

- **8 Specialized User Personas:** Complete dedicated UI modules and mathematical scoring algorithms for Health, Fitness, Beach/Surfing, Travel, Family/School, Agriculture, Commuting, and Event Planning.
- **Smart Auto-Blend Mode:** Automatically adapts recommendations throughout the user's daily lifecycle (Morning Fitness → Commute → School Pickup → Evening Leisure).
- **All-India States & Districts Directory Explorer:** Interactive explorer covering all 28 States and 8 Union Territories with instant 1-click station switching.
- **Universal Search & GPS Reverse Geocoding:** Converts exact coordinates into named towns/districts anywhere worldwide.
- **Multilingual Voice Briefing (TTS):** Speech synthesis delivering daily meteorological bulletins in English and हिन्दी (Hindi).
- **Simulated IMD Doppler Radar (DWR):** Live radar canvas sweep with reflectivity (dBZ) intensity color coding.

2. Matrix of the 8 Specialized Personas

Persona	Core Metrics & Features	Mathematical / Algorithmic Model	Actionable Output Example
 Health-Conscious	CPCB National AQI, PM2.5, PM10, NO ₂ , O ₃ , Pollen index, UV Sunburn timer	Indian AQI 6-tier piecewise linear interpolation; Asthma danger index via Humidity ($\text{H} > 80\%$) + PM2.5 ($\text{PM}_{2.5} > 60 \mu\text{g}/\text{m}^3$)	Moderate AQI "Wear N95 mask; Sunburn threshold ~25 mins"
 Outdoor Fitness	"Best Running Hours" curve, Sunrise/Sunset, WBGT heat index, Hydration rate	Suitability $S(0-100) = 100 - 4\Delta T - 0.5\Delta RH - 5\Delta UV - 1.5\Delta W$; Sweat rate $\text{SR} = 500 + 60\Delta T$ ml/hr	Peak Window: 06:00 AM "Score 92/100; Hydration: 680 ml/hr"
 Beach & Coastal	Coastal sea state, 24h Tide harmonic curve, Swell height & period, SST, UV reflection	Harmonic Tide $H = 1.6 + 1.2\sin(\frac{2\pi}{12.4}t + 0.8)$ m; INCOIS coastal safety thresholds	Calm to Moderate "High Tide: 02:45 PM (+2.8m); Ocean swell 1.2m"
 Smart Travel	Saved destinations, flight delays, railway fog alerts, AI Packing Advisor	Multi-variable rule matrix checking ΔT , rain sum, and UV max against clothing gear rules	AI Checklist "Pack raincoat & light fleece for Shimla (+14°C delta)"
 Parents & Family	School commute drop/pickup windows, Rain radar, Playground comfort index	Transit window segmentation (07-08:30 AM & 02-03:45 PM); Playground comfort $S(0-100)$	Commute Clear "Dry school transit; Playground safe after 04:30 PM"
 Agriculture (GKMS)	Topsoil (0-7cm) & Root-zone (7-28cm) moisture %, 7-day crop rain, Spraying window	Soil water saturation balance; Spraying suitability $(W < 15 \text{ ext}\{\text{km}/\text{h}\} \wedge \text{Pop} < 20\%)$	GKMS Advisory "Topsoil 34%; Hold irrigation; Spraying window open"
 Daily Commuter	Morning/Evening transit ratings, Road visibility meter (m), Waterlogging index	Atmospheric extinction coefficient visibility mapping; Urban drainage runoff capacity	Visibility: 3800m "Roads dry; Normal traffic across flyovers"
 Event Planner	14-day rain probability, Outdoor Thermal Comfort, Tent structural wind thresholds	Comfort $S = 100 - 3.5 T - 24 - 0.4(\text{Pop})$; Tent marquee gust rating ($\text{G} < 60 \text{ ext}\{\text{km}/\text{h}\}$)	Comfort: 88/100 "Best ceremony window: 06:30 PM - 10:30 PM"

3. System Architecture & File Structure

```
mausam-personalized-app/ |— server.py # Python 3 backend proxying Geocoding, Reverse-Geocoding &
Weather APIs |— static/ | |— index.html # Responsive UI with MoES/IMD styling, 8 personas &
All-India modal | |— css/ | | |— styles.css # Glassmorphism, mobile shell frame & animated
radar sweep | |— js/ | |— weatherService.js # Pan-India 36 States/UTs directory, live API &
AQI/UV banding | |— personaEngine.js # Algorithms, math models & widget renderers for all 8
personas | |— charts.js # Chart.js renderers (Tide curves, Workout scores, AQI, Agromet) | |—
packingAdvisor.js # AI forecast-aware packing checklist generator | |— voiceBriefing.js # Web
Speech API TTS voice engine (English & Hindi) | |— app.js # Global state orchestrator, search
autocomplete & modal manager |— README.md # System documentation and presentation guide
```

4. Detailed Line-by-Line Code Explanation

A. Backend Server (server.py)

- **Lines 1–11:** Imports Python standard libraries (`http.server`, `urllib`, `json`, `datetime`) and defines port 8000. Ensures 100% compatibility with zero third-party dependencies.
- **Lines 13–45 (do_GET router):** Parses URL path and dispatches incoming requests:
 - `/api/geocode`: Searches world and Indian cities with autocomplete candidates.
 - `/api/reverse-geocode`: Converts GPS coordinates into town and state names.
 - `/api/weather`: Bundles live forecast, air quality, marine, and soil moisture data.
 - `/api/health`: Health verification endpoint.
 - Default: Serves static frontend assets (HTML, CSS, JS).
- **Lines 47–53 (send_json_response):** Emits HTTP 200 with Content-Type: `application/json`, CORS header `Access-Control-Allow-Origin: *`, and a 300-second cache header.
- **Lines 55–90 (handle_geocode & handle_reverse_geocode):** Queries the high-speed geocoding endpoints with fallback to a comprehensive offline dataset of Indian district centers.
- **Lines 182–245 (handle_weather):** Queries Open-Meteo Synoptic Forecast, Air Quality Pollutants, and Coastal Marine APIs concurrently, merges them into a single payload, and attaches IMD provider metadata.

B. Meteorological & Geocoding Service (static/js/weatherService.js)

- **Lines 15–120 (indianStatesData):** Structured dictionary mapping all Indian States & UTs with district coordinates and coastal classification tags. Powers the **All States/Districts Explorer**.
- **Lines 122–140 (popularLocations):** Quick-access metro and zonal chips (New Delhi, Mumbai, Bengaluru, Puri, Shimla, Goa, Srinagar, etc.).
- **Lines 144–178 (searchLocation):** Fast search-as-you-type engine querying the backend geocoding proxy with instant client-side fallback across all Indian states.

- **Lines 180–190 (reverseGeocode):** Reverse geocodes device GPS coordinates into town and district labels.
- **Lines 232–260 (interpretWMO):** Converts WMO codes (0 to 99) into official Indian meteorological terminology (*Heavy Monsoon Rain, Thunderstorm with Squall, Dense Fog*).
- **Lines 262–282 (getIndianAQIBand & getUVBand):** Maps pollutant concentrations into CPCB 6-tier National AQI bands and UV radiation exposure categories.

C. Core Persona Engine (`static/js/personaEngine.js`)

- **Lines 72–115 (`render master dispatcher`):** Identifies the active persona and executes the matching layout renderer:
 - `renderHealthPersona`: Injects AQI gauge, sub-pollutants (PM2.5, PM10, NO₂, O₃), pollen risk, and clinical asthma advisories.
 - `renderFitnessPersona`: Injects WBGT heat stress gauge, hydration sweat loss (\$ml/hr\$), golden hour timings, and workout suitability timeline.
 - `renderBeachPersona`: Injects tide harmonics chart, high/low tide peak hours, wave heights (\$m\$), water temperature, and UV water reflection warnings.
 - `renderTravelPersona`: Injects transit delay radar, destination comparisons, and launches the AI Packing Advisor.
 - `renderFamilyPersona`: Injects morning drop-off (07-08:30 AM) and afternoon pickup (02-03:45 PM) school commute safety cards.
 - `renderAgriPersona`: Injects topsoil (0-7cm) and subsoil moisture %, spraying window indicators, and GKMS crop advisories.
 - `renderCommutePersona`: Injects fog/smog visibility distance meter (meters) and urban waterlogging risk index.
 - `renderEventPersona`: Injects 14-day rain probability curve, thermal comfort score, and temporary stage wind gust ratings.
 - `renderHybridPersona`: Weaves morning fitness, daytime transit, school commute, and evening leisure into an adaptive timeline.
- **Lines 847–865 (`initPersonaCharts`):** Binds Chart.js canvases after DOM updates.

D. Interactive Charts & Visualizations (`static/js/charts.js`)

- **Lines 14–68 (`renderAQIChart`):** Renders color-coded bar chart displaying individual concentrations of PM2.5, PM10, NO₂, O₃, SO₂, and CO.
- **Lines 70–145 (`renderWorkoutScoreChart`):** Computes hourly workout comfort scores (\$0–100\$) and plots them alongside ambient temperature curves.
- **Lines 147–205 (`renderTideCurveChart`):** Plots continuous 24-hour sinusoidal tide curves overlaid with swell heights (\$m\$).
- **Lines 207–265 (`renderAgriChart`):** Renders 7-day expected precipitation (\$mm\$) as blue bars overlaid with topsoil saturation percentage.
- **Lines 267–320 (`renderEventChart`):** Renders 10-day rain probability against the Outdoor Comfort Index.

E. AI Smart Packing Advisor (`static/js/packingAdvisor.js`)

- **Lines 27–75 (generatePackingList):** Inspects destination temperature minimums, rainfall projections, and UV maxima to dynamically generate clothing, gear, footwear, and health checklists.

F. Multilingual Voice Briefing (static/js/voiceBriefing.js)

- **Lines 7–90 (composeBulletin):** Generates localized weather, AQI, and persona advisories in English and हिन्दी (Hindi).
- **Lines 92–130 (speak & stop):** Connects to browser Web Speech API with hi-IN / en-IN voice synthesis.

G. Main Application Orchestrator (static/js/app.js)

- **Lines 17–40 (init):** Restores preferences from localStorage, initializes UI components, and boots weather data.
- **Lines 72–120 (setupDistrictExplorer):** Drives the All-India States & Districts directory modal.
- **Lines 122–145 (loadWeatherForLocation):** Fetches live data and triggers hero and persona updates.
- **Lines 342–480 (setupEventListeners):** Binds search bar with 250ms debounce, GPS reverse geocoding, voice modal, radar simulation, and mobile frame toggles.

Alignment with Ministry Standards

This system complies with guidelines from the **Ministry of Earth Sciences (MoES)**, **India Meteorological Department (IMD)**, **INCOIS (Marine)**, **CPCB (SAMEER AQI)**, and **Gramin Krishi Mausam Sewa (GKMS)**.